



R.V.R & J.C. COLLEGE OF ENGINEERING

Affiliated to Acharya Nagarjuna University (ANU)

SUMMER INTERNSHIP INTERNET OF THINGS

@LearnCorp

INCHARGE : Dr. T. Srilatha

Name: Rohith Kondapaneni

Branch: CSE(IOT)

Regd No: Y23CO029

Internship Name: IoT Summer Internship

Organization: LearnCorp

Mode: Online

What I Learned 🔧

Setting up IoT devices like ESP8266.ESP32

Using platforms like Blynk, Node-RED, Firebase

Collecting and visualizing sensor data

Creating dashboards for real-time monitoring

Building a smart home automation system

🎯 **Goal**

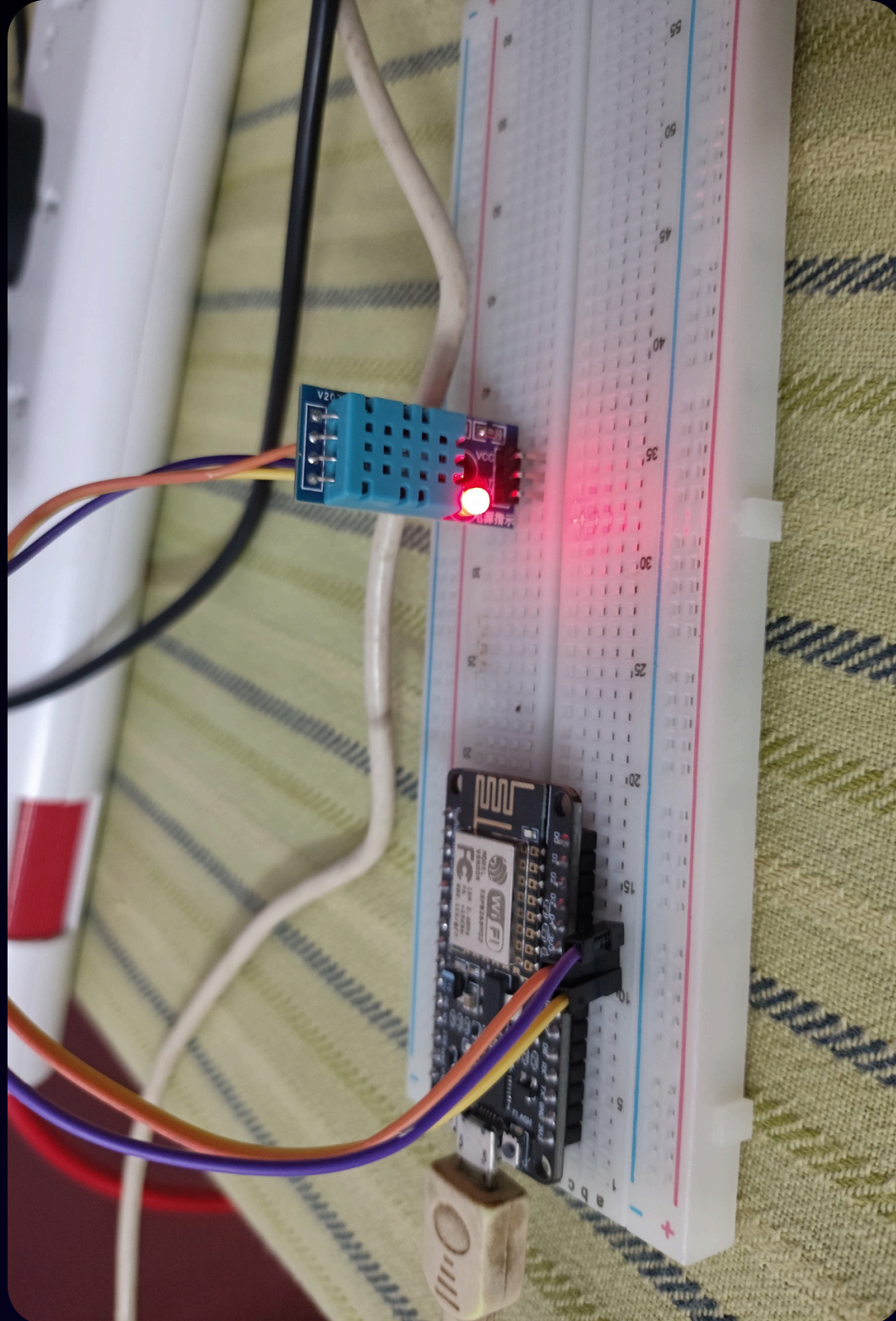
To gain practical experience in IoT by completing weekly projects.



KONDAPANENI ROHITH

S.NO: LC0125INOL082

Week 1 – No-Code IoT Setup



🔧 Task

Set up IoT devices using no-code platforms.

🔧 Tools Used

ESP8266, DHT11, PIR Sensor
Blynk, Node-RED, Cayenne

■ What I Did

- Connected sensors to ESP8266
- Used Blynk app to control and monitor data
- Created simple IoT flows with drag-and-drop tools

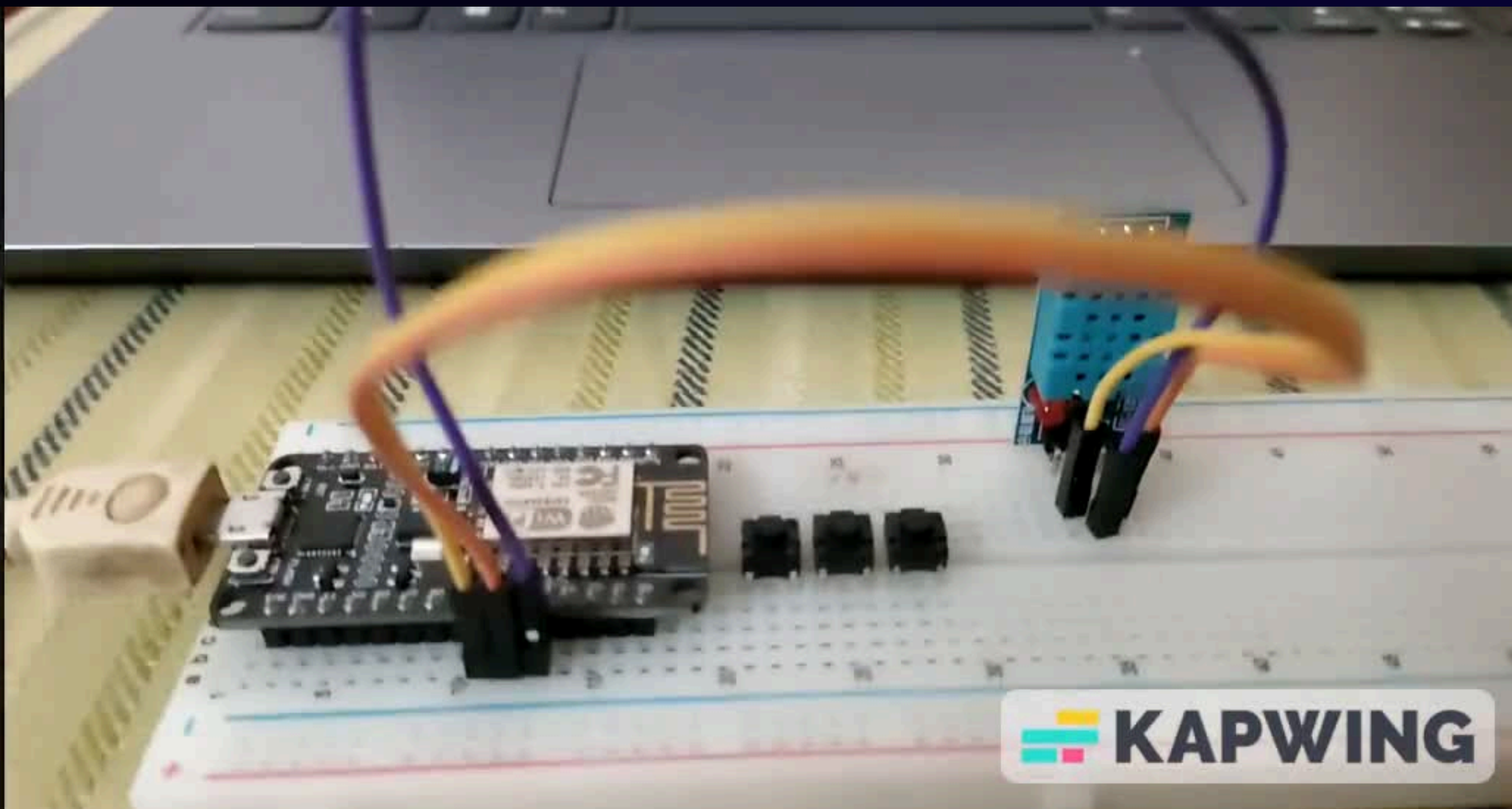
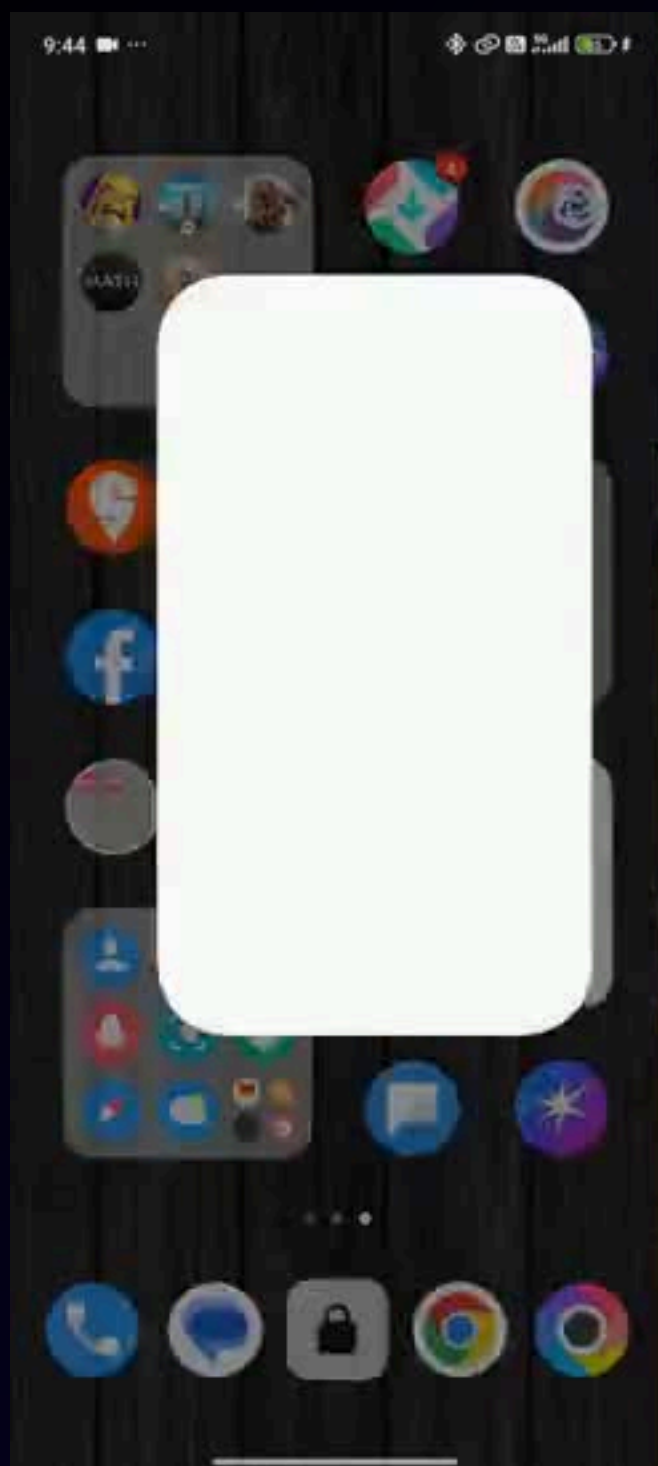
✅ Outcome

Built a basic IoT system without writing code.

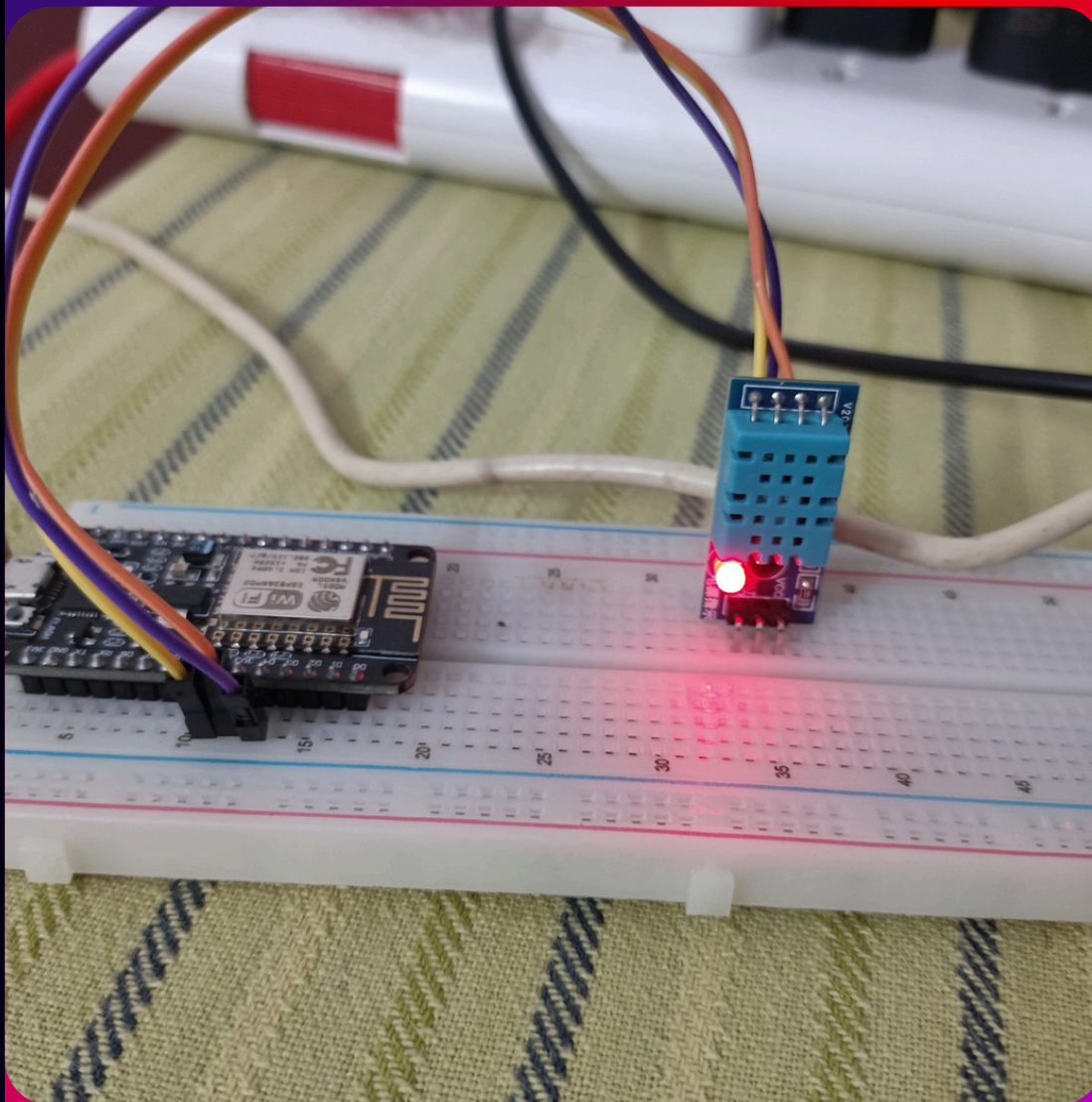
```

1  #define BLYNK_TEMPLATE_ID "TMPL3S-eM-FkG"
2  #define BLYNK_TEMPLATE_NAME "Quickstart Template"
3  #define BLYNK_AUTH_TOKEN "u9ARGQntbscK3Sf1bMi0Hkh1cUZiihtj"
4
5  #include <ESP8266WiFi.h>
6  #include <BlynkSimpleEsp8266.h>
7  #include <DHT.h>
8
9  char ssid[] = "IRONMAN";
10 char pass[] = "1234567890";
11
12 #define DHTPIN D4
13 #define DHTTYPE DHT11
14
15 DHT dht(DHTPIN, DHTTYPE);
16 BlynkTimer timer;
17
18 void sendSensor() {
19     float h = dht.readHumidity();
20     float t = dht.readTemperature();
21
22     if (isnan(h) || isnan(t)) {
23         return;
24     }
25
26     Blynk.virtualWrite(V0, t);
27     Blynk.virtualWrite(V1, h);
28 }
29
30 void setup() {
31     Serial.begin(9600);
32     Blynk.begin(BLYNK_AUTH_TOKEN, ssid, pass);
33     dht.begin();
34     timer.setInterval(2000L, sendSensor);
35 }
36
37 void loop() {
38     Blynk.run();
39     timer.run();
40 }
41

```

Week 2 – IoT Data Logging



🔧 Task

Log sensor data to cloud platforms in real time.

🔧 Tools Used

- ESP8266, DHT11 Sensor
- Firebase, Google Sheets
- Arduino IDE

🌐 What I Did

- Collected temperature & humidity data
- Sent data to Firebase and Google Sheets
- Created graphs to visualize live readings

✅ Outcome

Built a live data logger for remote monitoring.

```
1  #include <ESP8266WiFi.h>
2  #include <FirebaseESP8266.h>
3  #include <DHT.h>
4
5
6  #define WIFI_SSID "IRONMAN"
7  #define WIFI_PASSWORD "1234567890"
8
9  #define FIREBASE_HOST "nodemcu-dht11-38d6c-default-rtdb.firebaseio.com"
10 #define FIREBASE_AUTH "09sphIYrh2MGpxyVN11onG0zYJV5OLWJ0GjzsPgx"
11
12 #define DHTPIN D4
13 #define DHTTYPE DHT11
14
15 DHT dht(DHTPIN, DHTTYPE);
16
17 FirebaseData fbdo;
18 FirebaseAuth auth;
19 FirebaseConfig config;
20
```

```
21 void setup() {
22     Serial.begin(115200);
23     delay(1000);
24
25     Serial.print("Connecting to WiFi...");
26     WiFi.begin(WIFI_SSID, WIFI_PASSWORD);
27     while (WiFi.status() != WL_CONNECTED) {
28         delay(500);
29         Serial.print(".");
30     }
31     Serial.println(" connected!");
32
33     config.host = FIREBASE_HOST;
34     config.signer.tokens.legacy_token = FIREBASE_AUTH;
35
36     Firebase.begin(&config, &auth);
37     Firebase.reconnectWiFi(true);
38
39     dht.begin();
40 }
41
```

```
42 ✓ void loop() {
43     float humidity = dht.readHumidity();
44     float temperature = dht.readTemperature();
45
46 ✓ if (isnan(humidity) || isnan(temperature)) {
47     Serial.println("Failed to read from DHT sensor!");
48     delay(2000);
49     return;
50 }
51
52 Serial.printf("Temperature: %.2f °C, Humidity: %.2f %%\n", temperature, humidity);
53
54 ✓ if (Firebase.setFloat(fbdo, "/temperature", temperature)) {
55     Serial.println("Temperature uploaded successfully.");
56 ✓ } else {
57     Serial.print("Failed to upload temperature: ");
58     Serial.println(fbdo.errorReason());
59 }
60
61 ✓ if (Firebase.setFloat(fbdo, "/humidity", humidity)) {
62     Serial.println("Humidity uploaded successfully.");
63 ✓ } else {
64     Serial.print("Failed to upload humidity: ");
65     Serial.println(fbdo.errorReason());
66 }
67
68 delay(10000);
69 }
70
```


 **Firebase**

 Project Overview



Project shortcuts

 **Realtime Database**

What's new

 AI Logic

NEW

Product categories

Build

Run

Analytics

AI

 All products

Related development tools

 **Firebase Studio** 

Spark

No cost (\$0/month)

Upgrade

NEW

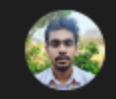


- Data
- Rules
- Backups
- Usage
-  Extensions

 `https://nodemcu-dht11-38d6c-default-rtdb.firebaseio.com`   

`https://nodemcu-dht11-38d6c-default-rtdb.firebaseio.com/
├── humidity: 80
└── temperature: 31.8`

 Database location: United States (us-central1)









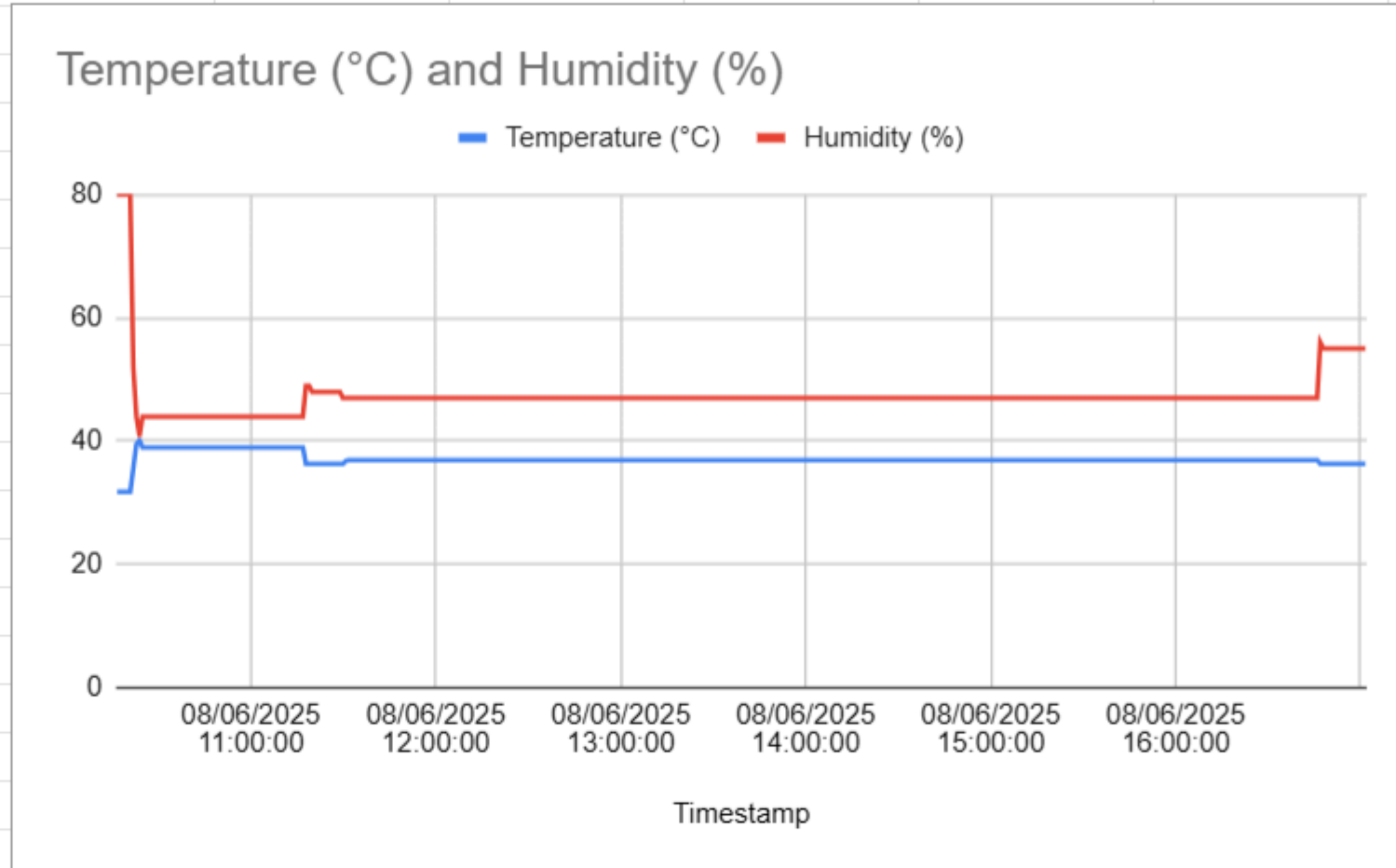


Temperature (°C) and Humidity (%)

Legend: Temperature (°C) (dark blue line), Humidity (%) (red line)

Timestamp	Temperature (°C)	Humidity (%)
07/06/2025 12:00:00	30	88
07/06/2025 16:00:00	29	86
07/06/2025 20:00:00	29	84
08/06/2025 00:00:00	29	83
08/06/2025 04:00:00	29	82

To exit full screen, press and hold **Esc**



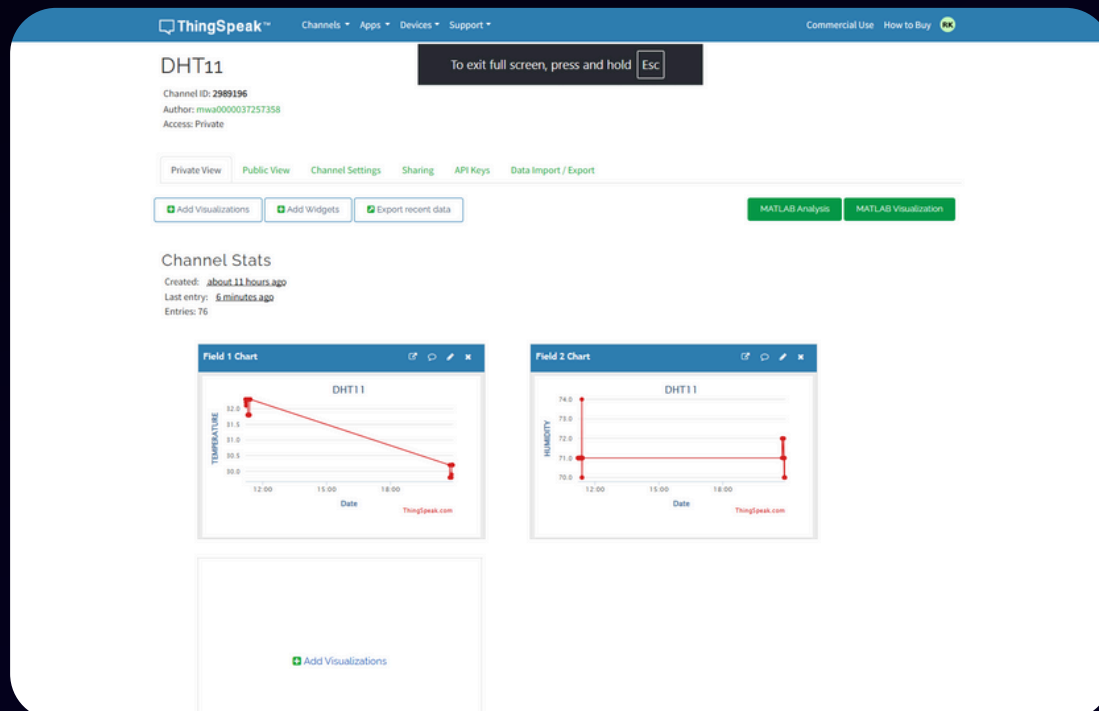
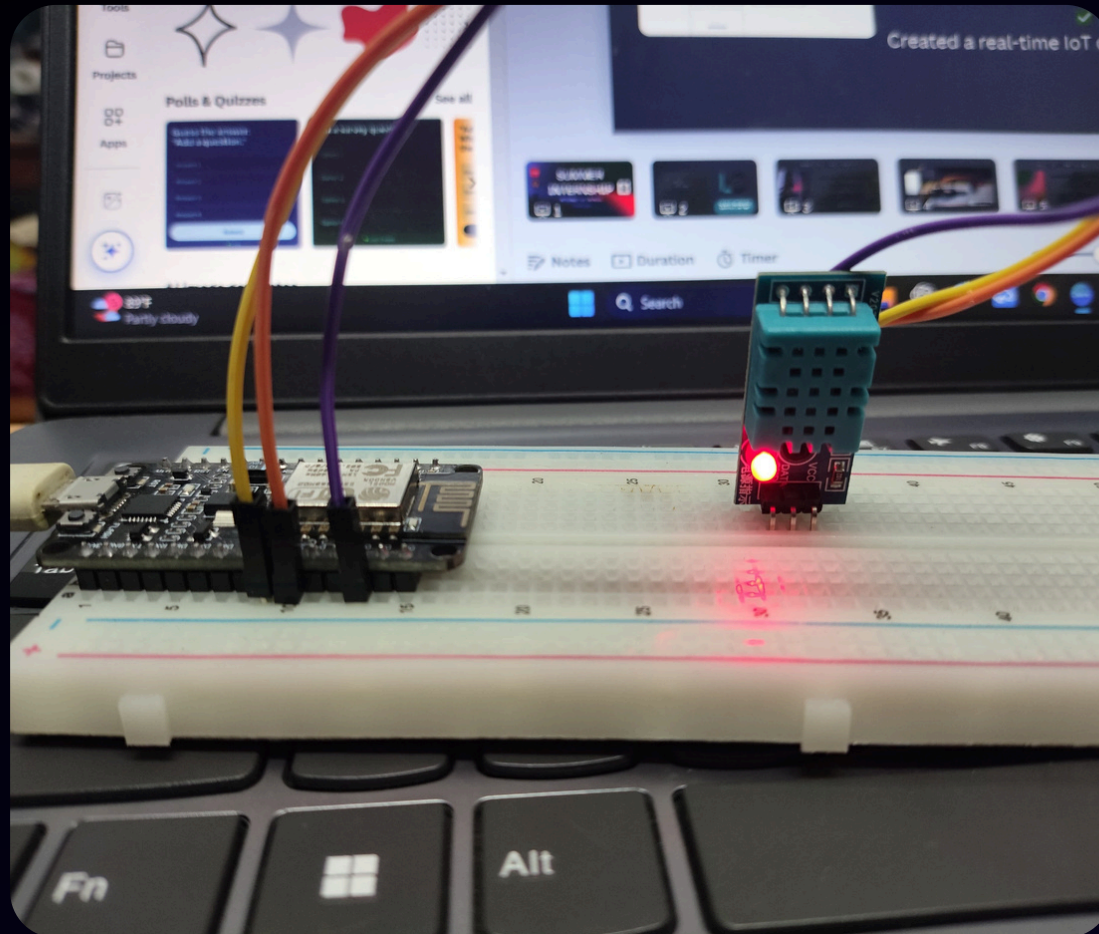
A1	Temperature (°C)														
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	Temperature (°C)	Humidity (%)	Timestamp												
2	31.8	80	08/06/2025 10:10												
3	31.8	80	08/06/2025 10:10												
4	31.8	80	08/06/2025 10:10												
5	31.8	80	08/06/2025 10:10												
6	31.8	80	08/06/2025 10:10												
7	31.8	80	08/06/2025 10:20												
8	35.6	52	08/06/2025 10:20												
9	39.5	44	08/06/2025 10:20												
10	40.1	41	08/06/2025 10:20												
11	40.1	41	08/06/2025 10:20												
12	39	44	08/06/2025 10:20												
13	39	44	08/06/2025 10:20												
14	39	44	08/06/2025 10:20												
15	39	44	08/06/2025 10:20												
16	39	44	08/06/2025 10:20												
17	39	44	08/06/2025 10:20												
18	39	44	08/06/2025 10:30												
19	39	44	08/06/2025 10:30												
20	39	44	08/06/2025 10:30												
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23	39	44	08/06/2025 10:30												
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25	39	44	08/06/2025 10:30												
26	39	44	08/06/2025 10:30												
27	39	44	08/06/2025 10:30												
28	39	44	08/06/2025 10:30												
29	39	44	08/06/2025 10:40												
30	39	44	08/06/2025 10:40												
31	39	44	08/06/2025 10:40												

Temperature (°C) and Humidity (%)

— Temperature (°C) — Humidity (%)

Timestamp	Temperature (°C)	Humidity (%)
08/06/2025 10:10	31.8	80
08/06/2025 10:20	35.6	52
08/06/2025 10:30	39	44
08/06/2025 10:40	39	44
08/06/2025 15:50	39	55
08/06/2025 16:00	39	44

Week 3 – Remote Monitoring Dashboard



Task

Monitor sensor data live from anywhere.

Tools Used

ESP8266, DHT11

ThingSpeak

Arduino IDE

What I Did

- Sent sensor data to cloud platforms
- Viewed live graphs and sensor values on mobile
- Set up alerts for temperature

Outcome

Created a real-time IoT dashboard with remote access.

```

1  #include <ESP8266WiFi.h>
2  #include "DHT.h"
3  #include "ThingSpeak.h"
4
5  #define DHTPIN D2
6  #define PIRPIN D5
7  #define DHTTYPE DHT11
8
9  const char* ssid = "IRONMAN";
10 const char* password = "1234567890";
11 WiFiClient client;
12
13 unsigned long myChannelNumber = 2989196;
14 const char * myWriteAPIKey = "SXLLB01VMCO44MUT";
15
16 DHT dht(DHTPIN, DHTTYPE);
17
18 void setup() {
19     Serial.begin(115200);
20     WiFi.begin(ssid, password);
21     dht.begin();
22     pinMode(PIRPIN, INPUT);
23
24     while (WiFi.status() != WL_CONNECTED) {
25         delay(1000);
26         Serial.println("Connecting to WiFi...");
27     }
28
29     Serial.println("Connected!");
30     ThingSpeak.begin(client);
31 }

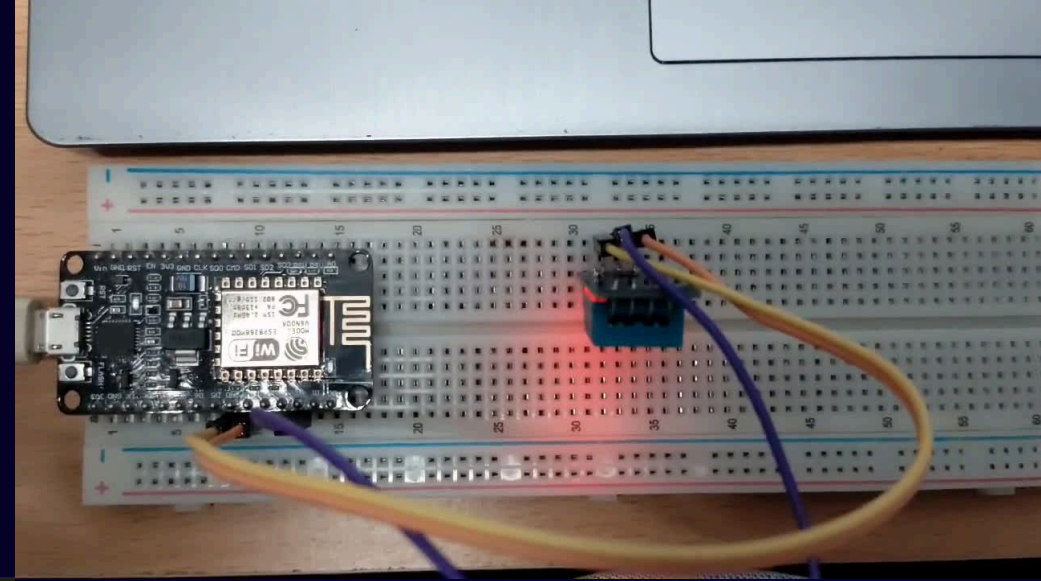
```

```

32
33 void loop() {
34     float t = dht.readTemperature();
35     float h = dht.readHumidity();
36     int motion = digitalRead(PIRPIN);
37
38     if (isnan(h) || isnan(t)) {
39         Serial.println("Failed to read from DHT sensor!");
40         return;
41     }
42
43     Serial.print("Temperature: "); Serial.println(t);
44     Serial.print("Humidity: "); Serial.println(h);
45     Serial.print("Motion: "); Serial.println(motion);
46
47     ThingSpeak.setField(1, t);
48     ThingSpeak.setField(2, h);
49     ThingSpeak.setField(3, motion);
50
51     ThingSpeak.writeFields(myChannelNumber, myWriteAPIKey);
52     delay(15000); // 15 sec delay (ThingSpeak limit)
53 }

```


WORKING VIDEOS



sketch_jun14a | Arduino IDE 2.3.6

File Edit Sketch Tools Help

NodeMCU 1.0 (ESP-12E Module) on COM12

sketch_jun14a.ino

```
21 dht.begin();
22 pinMode(PIR_PIN, INPUT);
23
24 while (WiFi.status() != WL_CONNECTED) {
25   delay(1000);
26   Serial.println("Connecting to WiFi...");
27 }
28
29 Serial.println("Connected!");
30 ThingSpeak.begin(client);
31 }
32
33 void loop() {
34   float t = dht.readTemperature();
35   float h = dht.readHumidity();
36   int motion = digitalRead(PIR_PIN);
37
38   if (isnan(h) || isnan(t)) {
39     Serial.println("Failed to read from DHT sensor!");
40     return;
41   }
42
43   Serial.print("Temperature: "); Serial.println(t);
44   Serial.print("Humidity: "); Serial.println(h);
45   Serial.print("Motion: "); Serial.println(motion);
46 }
```

Output

```
Writing at 0x00030000... (100 %)
Wrote 285552 bytes (209477 compressed) at 0x00000000 in 18.5 seconds (effective 123.2 kbit/s)...
Hash of data verified.

Leaving...
Hard resetting via RTS pin...
```

Building sketch

Ln 1, Col 1 NodeMCU 1.0 (ESP-12E Module) on COM12 2

Trending videos
First Time in Kyo...

Search

8:53:06 PM
6/14/2025